

NEUR-503 - Introduction to Computational Neuroscience

Modeling a Cortex-Basal Ganglia-Cerebellum Architecture Based on Doya (1999, 2000):
Implementation, Experimentation, and Critique

Nicolas Weill, McGill ID: 261323824

March 2026

Contents

1	Abstract	3
2	Introduction	3
2.1	Doya's foundational work	3
2.2	Why backpropagation is not physiologically accurate	4
3	Mathematical framework	4
3.1	The three learning paradigms	4
3.1.1	Reinforcement learning	4
3.1.2	Unsupervised learning	6
3.1.3	Supervised learning	7
4	Link between mathematical equations and biological reality	7
4.1	The cerebral cortex — unsupervised learning	7
4.1.1	Anatomy	7
4.1.2	Functioning, mathematical correspondence, and implementation	8
4.2	Basal ganglia — reinforcement learning	9
4.2.1	Anatomy	9
4.2.2	Functioning, mathematical correspondence, and implementation	10
4.3	Cerebellum — supervised learning	12
4.3.1	Anatomy	12
4.3.2	Functioning, mathematical correspondence, and implementation	13
5	Environment and Learning	15
5.1	Environment and task	15
5.2	Training loop (<code>train_cartpole_v2</code>)	16

6	Evaluation	16
6.1	Protocol	16
6.2	Results	17
6.3	Discussion	18
7	Critiques and improvements	18
7.1	Technical improvements	18
7.1.1	Cortex improvement	19
7.1.2	Switch to MLPs for the actor and critic (BasalGangliaACTrace)	19
7.2	Conceptual critiques and improvements	20
7.2.1	Implementation	21
8	Evaluation of technical and conceptual improvements	22
8.1	Protocol	22
8.2	Results	22
8.3	Discussion	24
9	Ablation experiments and simulation of neuropathologies	25
9.0.1	Cortical module ablation	25
9.0.2	Forward model ablation	25
9.0.3	Inverse model ablation	25
9.0.4	Simulation of Parkinson’s disease	26
9.0.5	Simulation of cerebellar ataxia	26
10	Learning a high-level task: Super Mario Bros	27
10.1	Architecture and implementation	27
10.2	Results and discussion	28
11	Conclusion and outlook	28

1 Abstract

This report presents the implementation, evaluation, and critical analysis of a computational architecture of the Cortex / Basal Ganglia / Cerebellum system inspired by the theoretical framework proposed by Kenji Doya in his works from 1999 and 2000.

The fundamental idea is that these three brain regions are not specialized by the type of information they process, but by the type of learning they perform: the cerebellum learns through supervised learning via climbing fibers; the basal ganglia through reinforcement learning via dopamine; and the cortex extracts internal representations through unsupervised learning.

The implementation relies on three Python modules: a CerebralCortex encoding the environment based on sparse coding principles, BasalGanglia Actor-Critic computing the temporal difference (TD) error, and a Cerebellum including a forward model and an inverse model. The architecture is evaluated on CartPole-v1, and then improved both technically and conceptually. The physiological relevance of the new model is then assessed through ablation experiments and disease simulations.

Finally, we show that by adding a vision module (CNN inspired by V1 cells), this architecture is capable of more complex tasks, notably playing Super Mario Bros on NES via the Gymnasium framework.

The results demonstrate the relevance of such a bio-inspired approach. However, although the excellent performance of the model allows it to be validated at the computational level, the model fails to reproduce certain physiological behaviors, notably tremors and rigidity in Parkinson’s disease.

2 Introduction

2.1 Doya’s foundational work

The question of the functional specialization of the cerebellum and basal ganglia has long been approached through the lens of motor control. The classical view assigned to the cerebellum a role of fine real-time correction, and to the basal ganglia a role of selection and inhibition of motor commands. This dichotomy has progressively been shown to be insufficient in light of the accumulation of functional imaging data showing the involvement of both structures in non-motor cognitive tasks: mental imagery, planning, language, sensory processing.

It is in this context that Kenji Doya proposed, in 1999 and then in 2000, a radically different unifying framework: rather than asking *what* each structure computes, he asks *how* each one learns. His thesis is based on a convergent anatomical observation: the cerebellum, basal ganglia, and cortex each exhibit a remarkably uniform and modular internal organization. If the local structure is identical throughout a given structure, then the learning algorithm implemented must also be universal and independent of the domain being processed.

The central thesis can be formulated as follows:

Structure	Paradigm	Learning signal
Cerebellum	Supervised	Vector error (climbing fibers)
Basal ganglia	Reinforcement	Scalar reward (dopamine)
Cerebral cortex	Unsupervised	Input statistics

However, these works have an important limitation: Doya essentially presents hypotheses and a mathematical model which, to our knowledge, has never been implemented in its entirety, probably due to the technological limitations of the

time.

Several reasons motivate an implementation of his theory in 2025–2026: first, computational power and deep learning libraries make it possible to test these ideas quickly on non-trivial tasks. Then, the structure proposed by Doya is now the subject of a scientific consensus [1]. It serves as a kind of solid framework on which to build and allows great freedom to attempt improvements both conceptual and technical. Finally, it makes it possible to further explore the question of physiologically plausible learning in a neural network, and to evaluate alternatives to backpropagation.

2.2 Why backpropagation is not physiologically accurate

Before getting into the heart of the matter, I wanted to include a short paragraph to explain why we aim to avoid backpropagation throughout this project. Indeed, although this rule is the current standard in deep learning, it runs into several major biological impossibilities:

- biological neurons in one layer do not know the synaptic weights of neurons in subsequent layers: they therefore cannot use them to compute their own gradient.
- biological plasticity is a local phenomenon: a neuron can only adjust its synapses based on the signals it directly receives (pre- and post-synaptic, or from global modulators). Backpropagation, on the other hand, requires global computations coordinated across the entire network.
- backpropagation imposes a strict alternation between a forward propagation phase and a backward propagation phase. The brain operates in an asynchronous and continuous manner, where perception and learning occur simultaneously without a global pause for gradient computation.

We will therefore explore alternatives to this algorithm throughout this report, including methods that are effective for handling a single layer such as Hebbian plasticity with lateral competition, as well as others that are effective at a multi-layer level: predictive coding, and the three-factor rule.

3 Mathematical framework

3.1 The three learning paradigms

This section formalizes the fundamental equations for each paradigm (reinforcement learning, supervised and unsupervised), as they appear in the original articles.

3.1.1 Reinforcement learning

The agent interacts with an environment described by a state $x(t) \in \mathbb{R}^n$. At each time step, the agent chooses an action $u(t)$ according to its policy $G(x(t))$. The state evolves as:

$$x(t+1) = F(x(t), u(t))$$

and a scalar reward $r(t+1) = R(x(t), u(t))$ is observed. We can therefore formalize a value function to maximize $V(x(t))$, expressing in a weighted way all future rewards:

$$V(x) = \mathbb{E}[r(t+1) + \gamma r(t+2) + \gamma^2 r(t+3) + \dots]$$

where $\gamma \in [0, 1]$ makes distant rewards less important: it is in some sense a measure of the network's impatience. This function can also be written in recursive form to make the Bellman equation appear:

$$V(x(t)) = r(t+1) + \gamma V(x(t+1)) \quad (\text{Eq. 1})$$

The fundamental algorithm for learning this value function V is TD learning, which seeks to minimize the temporal difference error:

$$\delta(t) = r(t) + \gamma V(x(t)) - V(x(t-1)) \quad (\text{Eq. 2})$$

- $V(x_{t-1})$ is what the agent thought the previous state was worth.
- $r(t) + \gamma V(x_t)$ is what it actually turns out to be worth: the reward actually received, plus the discounted value of the new state.
- $\delta(t)$ is the surprise — the gap between prediction and reality.

This signal δ is then used as an input signal by two agents: the Actor and the Critic.

The **Critic** aims to predict the value function as accurately as possible. As learning progresses and the context becomes more and more familiar to the critic, its prediction errors — the surprise — decrease:

$$\Delta v_j \propto \delta(t) \cdot x_j(t-1) \quad (\text{Eq. 3})$$

With Δv_j the change in weight v_j , that is, how much input j contributes to estimating the current value. Thus, if the surprise is positive ($\delta > 0$) and a certain input x_j was active ($x_j > 0$), then their product is positive $\rightarrow$ we increase $v_j \rightarrow$ next time we are in this state, we will estimate it more favorably. Thus the state will be considered positive by the value function and therefore by the policy, which will try more to reach it again; the opposite occurs when the surprise is negative.

The **Actor** aims to learn to select the best action $u(t)$ depending on the context. To avoid the agent getting stuck in a loop of suboptimal actions, Gaussian noise is added, allowing exploration of actions that the Actor would not have chosen spontaneously.

$$u(t) = g(Wx(t) + \xi(t)), \quad \xi \sim \mathcal{N}(0, \sigma^2 I) \quad (\text{Eq. 4})$$

where g is a non-linearity clipping to $[-1, 1]$.

$$\Delta w_{ij} \propto \delta(t) \cdot (u_i(t-1) - \bar{u}_i) \cdot x_j(t-1) \quad (\text{Eq. 5})$$

Here, Δw_{ij} represents the update of the weight linking input j to action i , i.e. the action to perform given the context.

The equation is the same as for the critic, but this time we add the term $(u_i(t-1) - \bar{u}_i)$, which is a measure of novelty: it is the difference between the action produced at the previous time step and the usual action. Thus:

- If an unusual deviation from the norm leads to a positive surprise ($\delta > 0$), the connection is strengthened to integrate this new action into the standard policy.
- If the deviation instead leads to a negative surprise ($\delta < 0$), the weight is adjusted in the opposite direction to penalize and avoid this action in the future.

3.1.2 Unsupervised learning

Unsupervised learning aims to extract a representation $y \in \mathbb{R}^m$ from the input $x \in \mathbb{R}^n$ ($m < n$) without an external signal. Doya adopts a criterion based on mutual information: maximize $I(x; y) = H(x) - H(x|y)$. Since $H(x)$ does not depend on the model, this amounts to minimizing the conditional entropy, thus making y as informative as possible about x , and therefore minimizing the reconstruction error.

The cost function combines this reconstruction error with a sparsity constraint:

$$E = \|x - W^\top y\|^2 + \lambda \sum_i |y_i| \quad (\text{Eq. 6})$$

where $\|x - W^\top y\|^2$ is the L2 reconstruction error ($W^\top y$ being an approximation of x), and $\lambda \sum_i |y_i|$ is a sparsity term that encourages neurons to remain inactive. In the non-linear case (neural network), $W^\top y$ is replaced by the output of a non-linear layer $f(Wy)$.

By computing $-\frac{\partial E}{\partial y}$, we obtain the relaxation dynamics that governs the inference of the representation:

$$\dot{y} \propto Wx - WW^\top y - \lambda \cdot \text{sign}(y) \quad (\text{Eq. 7})$$

The three terms have a clear interpretation: Wx is the excitation from the input (equivalent to a linear layer in an MLP), $WW^\top y$ is lateral competition (recurrent inhibition, to prevent several neurons from explaining the same thing), and $\lambda \cdot \text{sign}(y)$ is the sparsity pressure (L1). Since y appears on both sides of this differential equation, one must iterate until convergence: the system reaches a fixed point y^* that minimizes E for a given input x .

Similarly, $-\frac{\partial E}{\partial W}$ gives the Hebbian learning rule with synaptic decay, which updates the weights once y^* is reached:

$$\Delta W = y(x - W^\top y)^\top = yx^\top - yy^\top W \quad (\text{Eq. 8})$$

The term $yx^\top$ strengthens connections between active neurons and correlated inputs. The term $yy^\top W$ penalizes neurons that are too correlated with each other. Each neuron thus learns to explain x , but only the part that other neurons do not already explain. This rule is **entirely local**: no backpropagation.

3.1.3 Supervised learning

Supervised learning consists in learning a mapping $f : \mathbb{R}^n \rightarrow \mathbb{R}^p$ from pairs $(x, \hat{y})$ where $\hat{y}$ is the teaching signal. The update rule (Eq. 5 of Doya 1999) is:

$$\Delta w_{ij} \propto (\hat{y}_i - y_i) \cdot x_j \quad (\text{Eq. 9})$$

where $(\hat{y} - y)$ is the error—the climbing fiber signal—and x_j is the activity of parallel fiber j .

4 Link between mathematical equations and biological reality

The goal of this section is to show that the mathematical paradigms discussed in the previous section make it possible to explain the physiological functioning of the cortex, basal ganglia, and cerebellum. To do so, we will detail the anatomy and physiological functioning of each of these regions, which we will then relate to the mathematical equations presented in Part 3. Finally, we will present the implementation of the different modules.

4.1 The cerebral cortex — unsupervised learning

4.1.1 Anatomy

The neocortex exhibits an organization into six layers (see Fig. 1) with distinct cell types at each level. The superficial layers (I–III) mainly receive cortico-cortical projections, that is, connections originating from other cortical areas, while the deep layers (V–VI) send projections to subcortical structures (thalamus, basal ganglia). This architecture is universal: whether it is the visual, auditory, motor, or prefrontal cortex, the same structure reappears. It is precisely from this structural uniformity that Doya draws his conclusion: if the local organization is the same everywhere, the learning algorithm must also be universal.

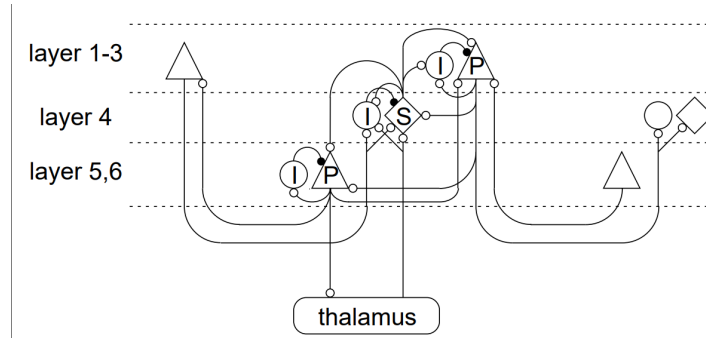


Figure 1: Diagram of the neural circuitry of the cerebral cortex (from Doya, 1999 [2]). P denotes pyramidal neurons; I denotes inhibitory interneurons. Open circles indicate excitatory connections, while filled circles represent inhibitory connections.

At the cellular level, two main classes of neurons can be distinguished: pyramidal neurons, which are excitatory and use glutamate as a neurotransmitter, and interneurons, which are inhibitory and use GABA. This excitation/inhibition opposition is fundamental, as it gives the cortex its ability to locally compete between competing representations — which, as we will see, corresponds directly to the lateral inhibition term in the relaxation dynamics equation.

4.1.2 Functioning, mathematical correspondence, and implementation

The central functional role of the cortex, in the framework of Doya (1999, 2000), is to construct a compressed and sparse representation of the sensory input x , denoted y . To achieve this, it optimizes the cost function E in two steps (see Fig. 2). The resulting representation is then transmitted to the basal ganglia and the cerebellum.

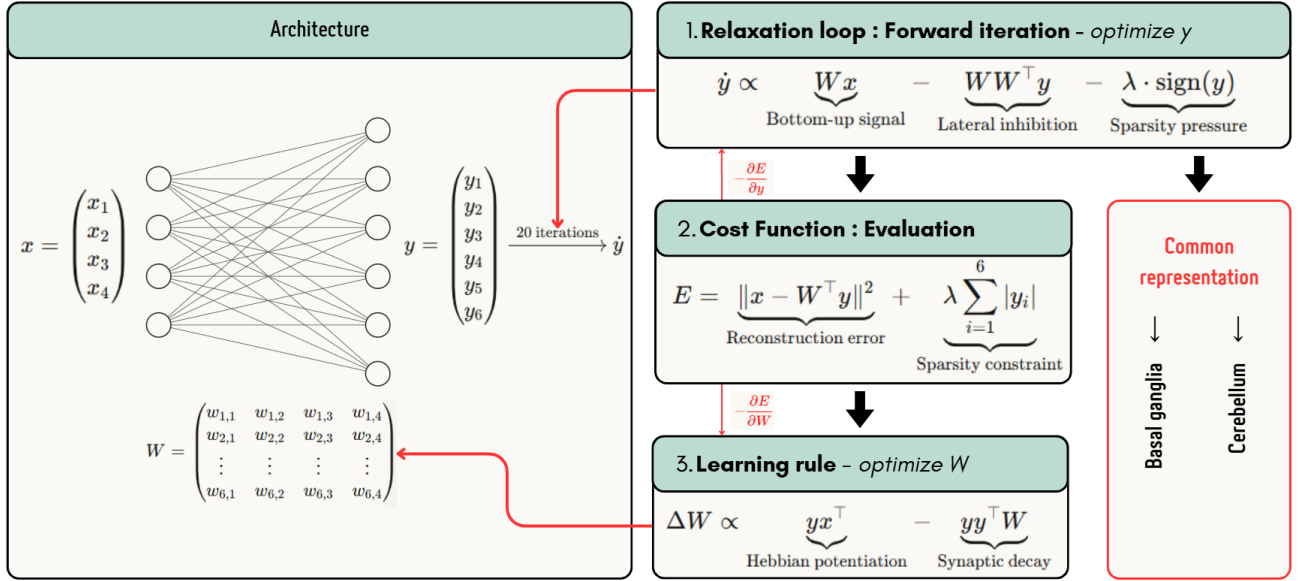


Figure 2: Architecture and mechanism of unsupervised learning in the cerebral cortex: relaxation loop (optimization of y , evaluation of the cost function E , and Hebbian update of the weights W).

Relaxation dynamics (Eq. 7) as recurrent neural activity. Biologically, the inference of y from x is not performed in a single feedforward pass: it arises from a recurrent dynamic within the cortex. Cortical neurons influence each other through feedback connections, and the activity of the network progressively converges toward a stable state. This mechanism is captured by the relaxation equation:

$$\dot{y} \propto Wx - WW^T y - \lambda \cdot \text{sign}(y) \quad (\text{Eq. 7})$$

Each of the three terms has a precise biological counterpart. The term Wx represents the feedforward inputs that convey sensory information to pyramidal neurons. These are the afferent excitatory synapses that depolarize target neurons in proportion to the correlation between their receptive field W_i and the input x .

The term $-WW^T y$ corresponds to lateral inhibition mediated by GABAergic interneurons, as discussed earlier. When a pyramidal neuron becomes active, it recruits inhibitory interneurons that suppress the activity of neighboring neurons with similar receptive fields, thereby enforcing a non-redundant representation.

Finally, the term $-\lambda \cdot \text{sign}(y)$ corresponds to the sparsity pressure. Physiologically, cortical neural networks exhibit sparse activity: in primary sensory areas, only a fraction of the neuronal population responds strongly to a given stimulus. This sparsity is energetically efficient, but it also maximizes the representational capacity of the network by making representations more separable. The L1 penalty on y is equivalent to a rectification threshold: a neuron becomes active only if its net excitation $Wx - WW^T y$ exceeds the cost λ .

The relaxation loop iterates until convergence to a fixed point y^* , reproducing the temporal dynamics observed in the cortex: neuronal activity in response to a stimulus does not instantaneously reach its steady state, but evolves over several

tens of milliseconds. In the code (`cerebral_cortex.py`, method `encode`), this loop is implemented by the following block:

Step 1: Encoding by relaxation (20 iterations).

```
1 Wx = self.W @ x
2 for _ in range(self.relaxation_steps): # 20 iterations
3     reconstruction = self.W.T @ y
4     recurrent = self.W @ reconstruction # WW'y: lateral competition
5     dy = Wx - recurrent - self.sparsity * np.sign(y)
6     y = y + self.relaxation_dt * dy # dt = 0.05
```

With $\text{self.W} @ x \equiv Wx$, $\text{self.W} @ \text{reconstruction} \equiv WW^\top y$, and $\text{sparsity} * \text{np.sign}(y) \equiv \lambda \text{sign}(y)$.

A small difference from Doya's equations: in the code, I introduced a row-wise normalization to make the weights more stable and facilitate convergence.

Hebbian rule (Eq. 8) as local synaptic plasticity. Once the representation y^* is stabilized, the synaptic weights W are updated according to the rule:

$$\Delta W = yx^\top - yy^\top W \quad (\text{Eq. 8})$$

This rule is entirely *local* (no backpropagation required): the modification of a synapse W_{ij} depends only on the pre-synaptic activity x_j (the input) and the post-synaptic activity y_i (the pyramidal neuron). The term $yx^\top$ is a classical Hebbian rule: it strengthens the connection between an active neuron and the inputs that activated it. The term $yy^\top W$, on the other hand, penalizes neurons that are too correlated with each other, thereby forcing each neuron to learn a distinct feature of x . In the code, this update is implemented as follows:

Step 2: Hebbian learning

```
1 error = x - self.W.T @ y
2 dW = np.outer(y, error) # Eq. 8: yx' - yy'W
3 self.W += self.lr * dW # lr = 0.005
4 self._normalize_weights() # unit norm per row
```

With $\text{np.outer}(y, \text{error}) \equiv \Delta W = yx^\top - yy^\top W$

Thus, the role of the cortex in Doya's architecture is twofold: it constructs a **shared representation** $y = f(x)$ for the basal ganglia and the cerebellum, and it performs an **adaptive dimensionality reduction** that simplifies the reinforcement learning problem of the basal ganglia — learning $V(y)$ on a sparse representation y is more efficient than learning $V(x)$ directly from raw sensory inputs.

4.2 Basal ganglia — reinforcement learning

4.2.1 Anatomy

The basal ganglia are a set of subcortical nuclei. The **striatum** (caudate nucleus and putamen) receives the main input from the cerebral cortex and is divided into two functionally distinct compartments (see Fig. 3):

- The **striosome** projects (inhibitorily) to the dopaminergic neurons of the **SNc** (*substantia nigra pars compacta*).

- The **matrix** is the starting point of the two main pathways: **the direct pathway** (which facilitates movement): Matrix $\rightarrow$ GPi/SN (inhibition) $\rightarrow$ Thalamus (released) $\rightarrow$ motor cortex; and **the indirect pathway** (which inhibits movement): Matrix $\rightarrow$ GPe $\rightarrow$ STN $\rightarrow$ GPi/SNr (reinforcement) $\rightarrow$ Thalamus (inhibited)

The SNc in turn sends **dopamine** to both compartments of the striatum.

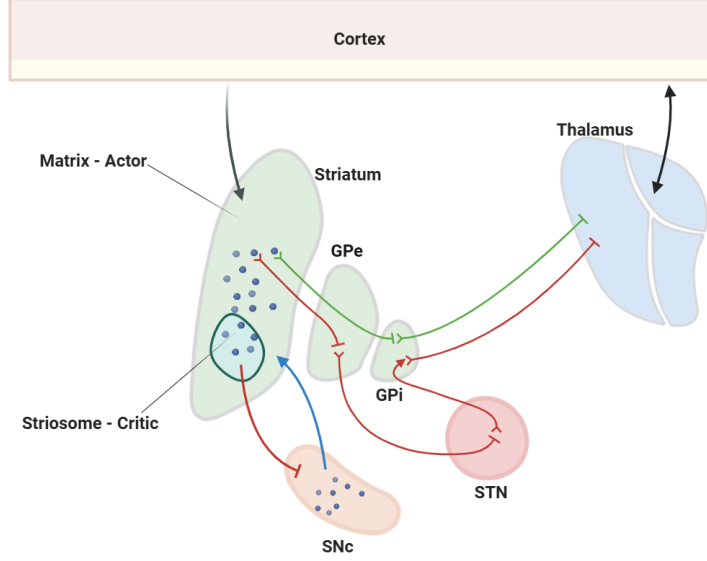


Figure 3: Diagram of neural circuit of the basal ganglia, made with biorender

4.2.2 Functioning, mathematical correspondence, and implementation

The striosome plays the role of the **Critic**: it computes the estimated value $V(x(t))$ of the current state. The matrix plays the role of the **Actor**: it selects the action $u(t)$ via the policy $G(x(t))$.

Dopaminergic neurons of the SNc encode the **temporal-difference error** (Eq. 2):

$$\delta(t) = r(t) + \gamma V(x(t)) - V(x(t-1))$$

This signal corresponds exactly to the recordings of Schultz et al. [11]: these neurons increase their activity when an *unexpected* reward occurs ($\delta > 0$), and decrease their activity below baseline when an expected reward does not occur ($\delta < 0$). Dopamine is therefore biologically the signal δ .

This signal asymmetrically modulates the plasticity of corticostriatal synapses: in the absence of dopamine, cortical co-activation induces LTD; in the presence of a dopaminergic burst, the same co-activation induces LTP. This corresponds to the learning rules of the Critic (Eq. 3) and the Actor (Eq. 5):

$$\Delta v_j \propto \delta(t) \cdot x_j(t-1) \quad (\text{Critic, striosome})$$

$$\Delta w_{ij} \propto \delta(t) \cdot (u_i(t-1) - \bar{u}_i) \cdot x_j(t-1) \quad (\text{Actor, matrix})$$

When the matrix selects an action ($\delta > 0$, positive reinforcement), it inhibits the SNr/GPi, releasing the thalamus to excite the cortex and trigger movement. The stochastic exploration of the Actor (Gaussian noise $\xi \sim \mathcal{N}(0, \sigma^2 I)$ in Eq. 4) corresponds biologically to fluctuations in the activity of striatal neurons that allow the system to explore new action sequences.

According to Doya [2], two types of architectures can be implemented:

Reactive action selection (Section 4.1 of Doya 1999). This is the most primitive architecture: stimulus $\rightarrow$ value $\rightarrow$ action $\rightarrow$ reward $\rightarrow$ update. The agent reacts to the current state **without simulating the future**. It is efficient for tasks with frequent and immediate rewards, but slow or unable to converge for tasks with sparse and delayed rewards.

```

1 # SNc (Eq. 2): TD error
2 next_value = 0.0 if done else self.value(next_state)
3 delta = reward + self.gamma * next_value - self._prev_value
4
5 # Striosome/Critic (Eq. 3)
6 self.v_weights += self.lr_critic * delta * self._prev_state
7
8 # Matrix/Actor (Eq. 5)
9 action_deviation = self._prev_action - self._prev_action_mean
10 self.w_weights += self.lr_actor * delta * np.outer(
11     action_deviation, self._prev_state)

```

The following table summarizes the correspondences between code and mathematical formulation:

Biological structure	Variable	Code
Striosome (Critic)	$V(x) = \mathbf{v}^\top x$	<code>v_weights</code>
Matrix (Actor)	$G(x) = g(Wx + \xi)$	<code>w_weights</code>
SNc (dopamine)	$\delta = r + \gamma V' - V$	<code>delta</code>
Corticostriatal LTP	$\delta > 0 \Rightarrow \Delta w > 0$	<code>+= lr * delta * ...</code>
Corticostriatal LTD	$\delta < 0 \Rightarrow \Delta w < 0$	same equation

Table 1: Biology–implementation correspondence, basal ganglia

Predictive action selection (Section 4.2 of Doya 1999). To overcome the limitations of the reactive mode, Doya proposes using the cerebellar forward model to simulate one step ahead (the biological plausibility of this approach was demonstrated by Wolpert *et al.* in 1998 [13]). The agent generates n candidate actions, predicts the next state for each via $F(s_t, u^{(k)})$, evaluates them with the Critic, and executes the optimal action:

$$u^* = \arg \max_k [R(s_t, u^{(k)}) + \gamma V(F(s_t, u^{(k)}))]$$

. During training, noise is added to the optimal action to facilitate exploration.

The definition and choice of learning strategies (reactive or predictive) are implemented in the training files, in the methods `_select_action_reactive` and `_select_action_predictive`.

Figure 4 summarizes the information flow and functioning of the basal ganglia implementation:

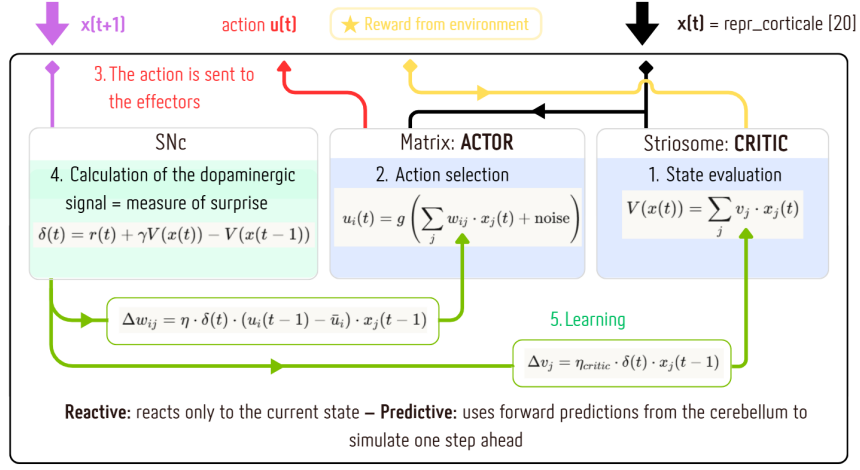


Figure 4: Representation of the information flow and functioning of the basal ganglia implementation.

4.3 Cerebellum — supervised learning

4.3.1 Anatomy

The structure of the cerebellum is considered to be one of the most uniform in the brain: all its subregions exhibit the same almost entirely feedforward organization. This uniformity is Doya's central argument for postulating a single learning algorithm (supervised learning) independent of the domain (medial motor, lateral cognitive).

Two main input pathways can be identified (see Fig. 5):

- **Mossy fibers** transport afferent sensory signals and cortical efferences. Their signal constitutes the input $x(t)$ of the model.
- **Climbing fibers**, originating from the inferior olive, converge onto Purkinje cells: each Purkinje cell receives only one climbing fiber, but about 200,000 inputs from parallel fibers.

Mossy fibers first relay their signal through **granule cells**, whose axons — the parallel fibers — reach the Purkinje cells. These granule cells are present in very large numbers (estimated between 10^4 and 10^6) and allow an expansion of the representation, projecting the input into a very high-dimensional space. The **deep cerebellar nuclei** then receive inhibition from Purkinje cells and direct excitation from mossy fibers, and produce the final output command.

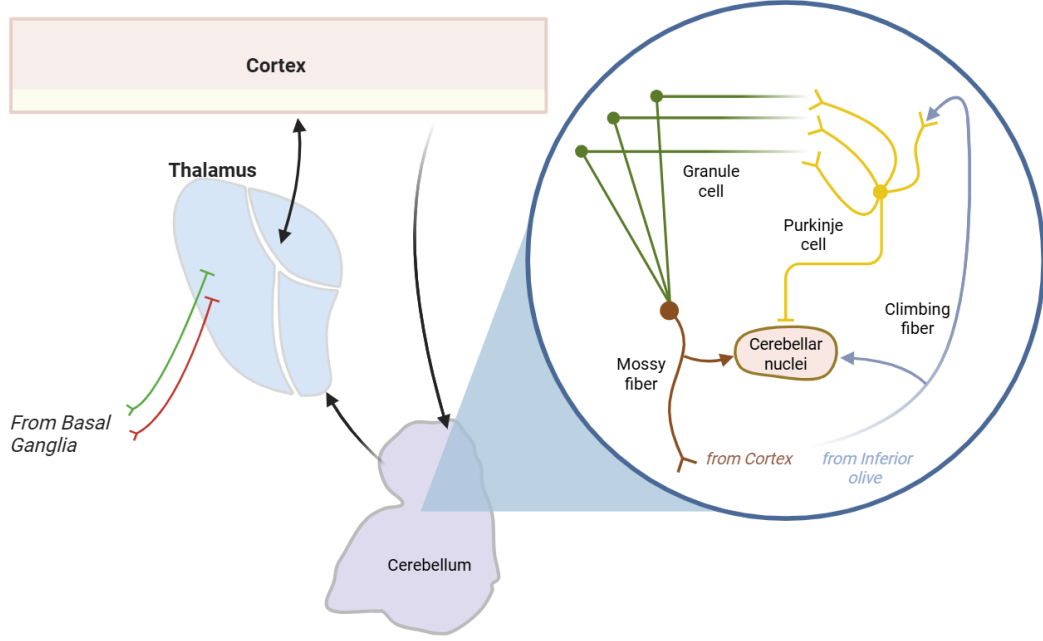


Figure 5: Schematic of the cerebellar circuit and its extra-cerebellar projections. Created with BioRender.com.

Doya notes that the cerebellum is divided into microzones of about 3,000 Purkinje cells each, each processing a distinct control subproblem. This internal modularity is consistent with the idea that the cerebellum implements multiple internal models in parallel.

4.3.2 Functioning, mathematical correspondence, and implementation

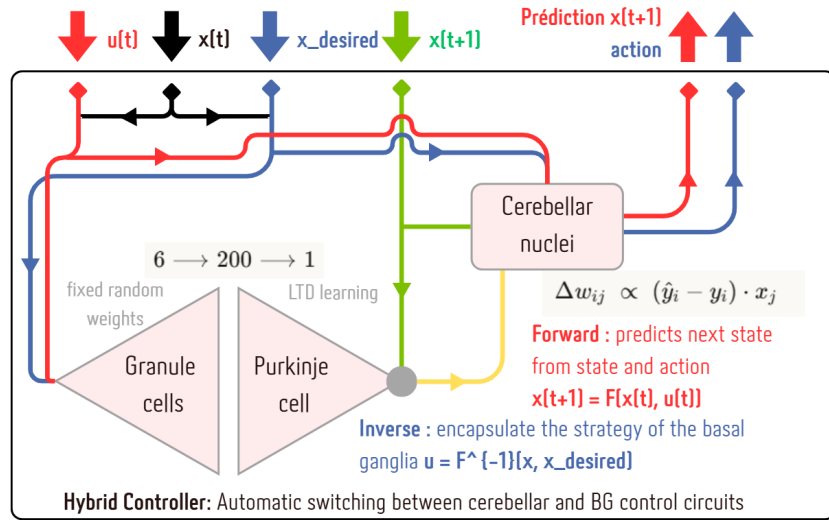


Figure 6: Computational architecture of the hybrid cerebellar controller

The implementation relies on an encoder representing granule cells and a decoder representing a Purkinje cell and its different input dendrites. Learning occurs only at the level of the Purkinje cell, which is consistent with biological observations: plasticity in the cerebellum is primarily concentrated at the synapses between parallel fibers and Purkinje cells, while the granule layer mainly performs a fixed expansion of the input space.

Granule cells project the input $x \in \mathbb{R}^n$ into a much higher-dimensional space $N_g \gg n$ through fixed and random weights, producing the activations of the parallel fibers $\phi(x) \in \mathbb{R}^{N_g}$. As noted earlier, this layer is never updated:

```

1 # Fixed weights, initialized once
2 self.gc_weights = np.random.randn(n_granule, input_dim)
3 # Sparse ReLU activation
4 gc = np.maximum(0, self.gc_weights @ x + self.gc_bias)

```

Purkinje cells and LTD (Eq. 5). Purkinje cells compute a linear combination of parallel fibers $y_i = \sum_j w_{ij} \cdot \phi_j(x)$, and their weights are updated using Doya's rule (Eq. 5):

$$\Delta w_{ij} \propto (\hat{y}_i - y_i) \cdot \phi_j(x) \quad (1)$$

where $(\hat{y}_i - y_i)$ is the error carried by the climbing fiber. In the code, this translates to:

```

1 prediction = self.pc_weights @ gc + self.pc_bias # Purkinje output
2 climbing_error = target - prediction # climbing fiber signal
3
4 # LTD of parallel fiber -> Purkinje synapses (Eq. 5)
5 self.pc_weights += self.lr * np.outer(climbing_error, gc)
6 self.pc_bias += self.lr * climbing_error

```

This rule is entirely local: each Δw_{ij} depends only on the post-synaptic error $(\hat{y}_i - y_i)$ and the pre-synaptic activity $\phi_j(x)$, without any global information.

Forward model and inverse model. In the implementation, the cerebellum is composed of two subclasses: **ForwardModel** and **InverseModel**, corresponding to two functional subdivisions of the lateral cerebellum [2].

ForwardModel implements the **predictive role** of the lateral cerebellum: it receives an efference copy of the motor command $u(t)$ along with the state $x(t)$, and predicts the sensory consequences $x(t+1)$ before peripheral feedback arrives. This enables two critical functions:

- Allow the cortex to correct an action in real time via the prediction error signal.
- Allow the basal ganglia to plan a strategy (**predictive strategy**) rather than simply relying on the present state (**reactive strategy**).

InverseModel implements the **motor encapsulation** role: it learns to reproduce the policy of the basal ganglia, transforming a desired state x_{desired} directly into a command u , automatically. Doya [2] describes this mechanism as the progressive transfer of control from the basal ganglia (slow and explicit system) to the cerebellum (fast and automatic system) over the course of learning. This process corresponds biologically to the **automation of motor skills**.

Only their inputs and predictions differ, which illustrates the architectural uniformity of the cerebellum emphasized by Doya:

```

1 # Forward model: F(x(t), u(t)) -> x(t+1)
2 x_fwd = np.concatenate([state, action]) # input: state + action
3 self.learn(x_fwd, next_state) # target: next observed state
4
5 # Inverse model: F^{-1}(x, x_desired) -> u

```

```

6 x_inv = np.concatenate([state, target])    # input: state + desired state
7 self.learn(x_inv, correct_action)          # target: effective action

```

Online correction by the CerebellarCorrector. The prediction error of the forward model generates a real-time corrective signal, added to the base action via a finite-difference approximation of the inverse Jacobian:

$$u_{\text{corr}} = u_{\text{base}} + \alpha \cdot J^{\top}(x_{\text{obs}} - x_{\text{predicted}}), \quad J_{ij} \approx \frac{F_i(x, u + \varepsilon e_j) - F_i(x, u)}{\varepsilon} \quad (2)$$

```

1 state_error = observed_state - forward_model.predict_next_state(prev_state, prev_action)
2
3 # Jacobian approximated by finite differences
4 for j in range(action_dim):
5     perturbed[j] += eps
6     jacobian[:, j] = (forward_model.predict_next_state(prev_state, perturbed) - predicted_state) / eps
7
8 correction = self.correction_gain * (jacobian.T @ state_error)
9 corrected_action = base_action + correction

```

Hybrid controller. This feature is simply mentioned by Doya, but he does not formalize it mathematically. The HybridController uses prediction error as a surprise detector: if $\|x_{\text{obs}} - x_{\text{predicted}}\|$ exceeds an adaptive threshold (moving average ± 2 standard deviations), control switches back to the basal ganglia. Otherwise, the cerebellar inverse model takes over, being faster and less costly.

```

1 if prediction_error > self.surprise_threshold:
2     # BG deliberation (slow, costly but flexible)
3     action = basal_ganglia.policy(bg_state, explore=False)
4 else:
5     # Fast CB (learned automatism)
6     action = inverse_model.compute_action(raw_state, target)
7
8 # Final refinement by the corrector
9 action, _ = corrector.correct(observed_state, action)

```

5 Environment and Learning

5.1 Environment and task

The CartPole-v1 task from the gymnasium framework by OpenAI has been implemented. It is a classic of *reinforcement learning*: the goal is to balance a pole on a cart by only being able to move it to the left or to the right. The state is a vector of dimension 4: $[pos_{\text{cart}}, vel_{\text{cart}}, \theta, \dot{\theta}]$, and the action is binary (0: left or 1: right). The reward is +1 for each time step during which the pole remains balanced.

5.2 Training loop (train_cartpole_v2)

At each time step t , the following operations take place:

- **Cortical encoding.** The cortex encodes the raw state $x(t)$ into a compressed representation $y(t)$ while updating its weights (Eq. 8):

```
1 cortex_repr, cx_err = cortex.encode_and_learn(raw_state)
```

- **Action selection.** In *reactive* mode, the basal ganglia directly select an action from the current state. In *predictive* mode, the forward model evaluates each candidate action (from the cerebellum `ForwardModel`) and selects the one that maximizes $\delta^* = \gamma V(x^*(t+1)) - V(x(t))$:

```
1 # Reactive
2 action = basal_ganglia.policy(bg_state, explore=True)
3
4 # Predictive (CartPole only)
5 action, discrete = _select_action_predictive(
6     basal_ganglia, forward_model, cortex, raw_state, bg_state
7 )
```

- **Interaction with the environment.** The action is applied, and the environment returns the next state $x(t+1)$ and the reward $r(t)$.
- **Reinforcement learning (basal ganglia).** The TD error $\delta(t)$ is computed and used to update the critic (striosome, Eq. 3) and the actor (matrix, Eq. 5):

```
1 delta = basal_ganglia.learn(reward, next_bg_state, done)
```

- **Supervised learning (cerebellum).** The forward model learns the observed state transition $(x(t), u(t)) \rightarrow x(t+1)$, and the inverse model encapsulates the action that has just been executed :

```
1 fwd_err = forward_model.learn_transition(raw_state, action, raw_next)
2 inv_err = inverse_model.learn_action(raw_state, target_state, action)
```

6 Evaluation

6.1 Protocol

The protocol is carried out in two phases:

1. **Training.** The model is trained in reactive mode, then in predictive mode, with a fixed seed (`seed=42`) for reproducibility. `ReachingEnv` is trained for 1,000 episodes per mode, and `CartPole` for 1,500 episodes per mode.
2. **Test phase.** After each training, the four control strategies are evaluated without exploration over 50 independent trials (`seed=999`):

- **BG (Basal Ganglia)** — policy of the basal ganglia alone;

- **Predictive** — selection by the BG using predictions from the cerebellum `ForwardModel`;
- **Inverse** — cerebellum `InverseModel` alone;
- **Hybrid** — automatic switching between BG control and `InverseModel` using the `HybridController` method.

Note: To evaluate the quality of the cortex representation, the architecture was also trained by concatenating to the cortex output the features from the environment and manually engineered features:

```
1 features = np.concatenate([
2     obs,                                # 4: raw observation
3     [abs(angle)],                        # 1: deviation magnitude
4     [angle * ang_vel],                  # 1: falling indicator
5     [np.sign(cart_pos)],                 # 1: side of center
6     [cart_vel * np.sign(angle)],        # 1: velocity alignment with tilt
7     cortex_repr,                         # repr_dim: learned features
8 ])
```

Cela permet de dissocier la contribution propre du cortex non supervisé de celle des features géométriques explicites, et d'évaluer si la représentation apprise apporte une valeur ajoutée au-delà d'un encodage manuel.

6.2 Results

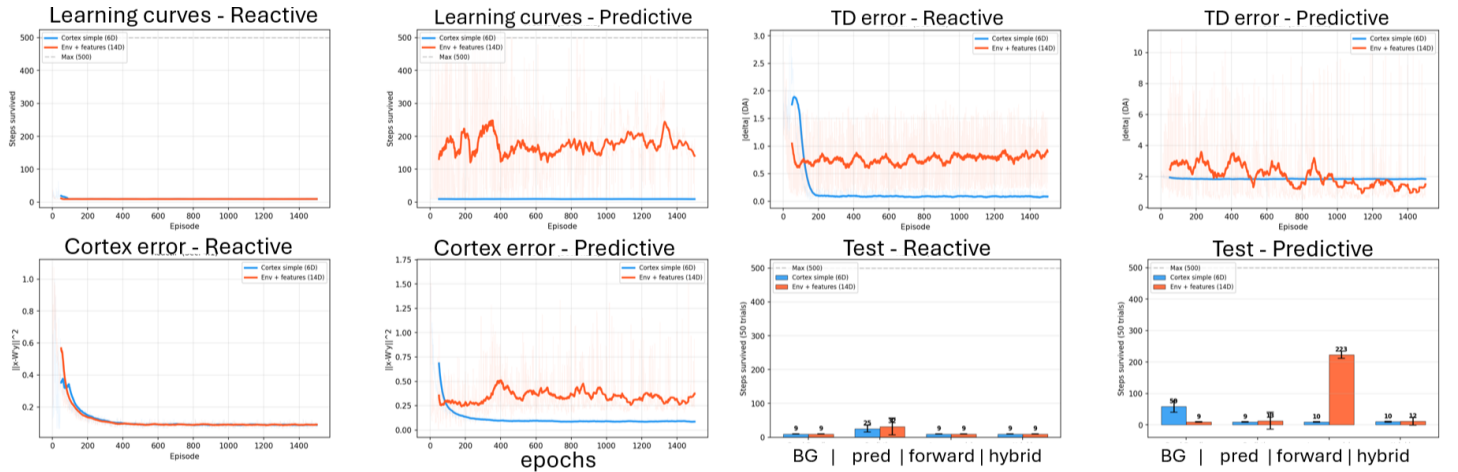


Figure 7: Benchmark — Doya (1999) Architecture on CartPole-v1: Cortex-only Representation (6D) vs. Manually Augmented State (14D), Reactive and Predictive Modes

The results are presented in Figure 7. Two main observations emerge:

First, the simple cortex (6D, blue) fails to learn in both modes: the reward remains zero over all 1,500 episodes, the TD error converges to zero due to the absence of significant new activity, and all test strategies plateau at around 9 steps. All test strategies plateau at around 9 steps. The cortical reconstruction error does decrease, which confirms that the unsupervised module is indeed learning a representation—but this representation is not sufficiently informative for the basal ganglia to extract a viable policy.

Second, adding manual features (14D, orange) unlocks learning in predictive mode: the reward curve oscillates between 100 and 250 steps, and the TD error stabilizes around 2. In reactive mode, even with the features, learning remains very weak. The most notable result concerns the *Inverse model* strategy in predictive mode, which reaches **223 steps** in testing—by far the best observed performance—compared to only 10 steps for the simple cortex under the same conditions.

Finally, it can be observed that when learning takes off (i.e., in predictive mode with manual features), the reward curve rises rapidly during the first 10 episodes, then stagnates on a plateau without notable progression. This behavior is characteristic of convergence to a *local minimum*.

6.3 Discussion

Several conclusions can be drawn from these results

Insufficiency of the cortex alone. The complete failure of the 6D cortex highlights a fundamental limitation: in CartPole, the critical parameter (imminent pole fall) is encoded by a nonlinear relationship between θ and $\dot{\theta}$.

- **Problem #1:** The cortex (4 neurons $\rightarrow$ 6 neurons, without activation function) simply does not provide enough nonlinearity and complexity to represent this problem; moreover, features related to velocity require access to previous frames to emerge.
- **Problem #2:** The cortex learns to reconstruct x in a compact way, but has no signal indicating which features are relevant for the task. The representation y is therefore statistically faithful to the input, without being functionally useful for control.

Limitations of the current basal ganglia. Moreover, even with augmented nonlinear features, the basal ganglia converge to a local minimum:

- **Problem #3:** The actor and the critic rely on purely linear functions, which is insufficient to approximate the nonlinear CartPole problem. Even if manual features introduce some nonlinearity upstream, the approximation capacity remains bounded by this linearity.

Nevertheless, one can still observe the good performance of the inverse model in predictive mode (223 steps). This result is explained by the quality of the teaching signal received by the cerebellum: in predictive mode, the executed actions are those that the forward model considers most promising, which provides the inverse model with higher-quality action examples to imitate. In reactive mode, the inverse model encapsulates a basal ganglia policy that has not itself learned to control the task, making its teaching signal unusable. This illustrates the cascade dependency of the three modules: the quality of cerebellar encapsulation is bounded by the quality of the policy it aims to reproduce.

7 Critiques and improvements

7.1 Technical improvements

The technical improvements therefore focus on two main axes:

- 1. **Improving the cortex architecture**, to be able to extract higher-level features $\rightarrow$ *to solve problem #1*
- 2. **Moving from linear actor and critic to two MLPs**, which allows approximating arbitrarily complex value functions and policies, and introducing a more biologically plausible learning rule via eligibility traces $\rightarrow$ *to solve problem #3*

7.1.1 Cortex improvement

The improvements described below are implemented in the class `CerebralCortexImproved` in the file `cerebral_cortex_improved.py`.

Temporal frame stacking (*frame stacking*). The first modification consists in concatenating sensory signals at time t with those from $t-1, \dots, t-n$, with $n=3$. This allows the cortex to implicitly access features derived from state derivatives: the difference between two consecutive frames approximates velocity, the difference of velocity approximates acceleration, and so on. In the case of CartPole, this is particularly useful: $\dot{\theta}$ is directly provided, but the trend of $\dot{\theta}$ (i.e., angular acceleration) is not observable in a single time step.

Biologically, this mechanism corresponds to **working memory cells** discovered by Fuster in the prefrontal cortex [5]: these neurons maintain persistent activity (*persistent firing*) throughout the duration of a memory delay, via NMDA-dependent recurrent re-excitation loops, allowing a cortical region to integrate the recent history of its inputs without an explicit memory mechanism.

Improvement of the cortex architecture The next step is to move to a cortex architecture capable of taking into account our new input. Thus, we move from a relatively simple architecture (4D $\rightarrow$ 6D) to an architecture including a hidden layer (16D $\rightarrow$ 32D $\rightarrow$ 16D). The weights of this hidden layer are fixed at initialization and are never updated (I was inspired by the logic of granule cell weights = kernel expansion). This is one of the constraints inherent to the use of a hidden layer—see the section. Still, a sufficiently large random projection is often enough to capture most of the complexity without requiring the computational cost (and the risk of overfitting) associated with training by backpropagation.

Finally, to introduce nonlinearity into the system, the inputs of the hidden layer are activated by the nonlinear tanh function, which biologically corresponds to nonlinear dendritic integrations.

7.1.2 Switch to MLPs for the actor and critic (BasalGangliaACTrace)

The fundamental limitation of the original actor-critic is its linearity: $V(x) = \mathbf{v}^\top x$ and $u = g(Wx + \xi)$ can only represent hyperplanar decision boundaries. The `BasalGangliaACTrace` class replaces these linear functions with two MLPs (128D $\rightarrow$ 64D, ReLU), one for the critic and one for the actor. This architecture is capable of approximating any continuous function, allowing the critic to learn a nonlinear value function and the actor to represent policies with complex decision boundaries. Biologically, these neurons correspond to neurons in the striosomal and matrix compartments, whose dendrites perform nonlinear integration of cortical inputs.

Eligibility trace and three-factor rule. However, to update MLPs, we can no longer use our Hebbian rule $\Delta w \propto \delta \cdot (u \otimes x)$. Indeed, this rule is purely local and cannot coordinate changes across multiple layers to minimize a distant error. We therefore turn to the three-factor rule (Frémaux & Gerstner 2016), (see paragraph 2.2). This rule overcomes the limitations of the Hebbian rule by using an eligibility trace that marks synapses that have been recently active (to identify which ones contributed to the previous action, so that their weights can be updated based on the dopaminergic signal).

The three-factor rule can be expressed in our case by the following formula:

$$\Delta W(t) = \alpha \cdot \delta_{\text{centered}}(t) \cdot e(t-1), \quad e(t) = y_t \otimes (x_t - W^\top y_t)$$

$$\Delta W(t) = \alpha \cdot \underbrace{\delta(t)}_{\text{dopamine (SNc)}} \cdot \underbrace{e(t)}_{\text{eligibility trace}}$$

where the eligibility trace is updated at each step:

$$e(t) = \gamma \lambda \cdot e(t-1) + \nabla_W \log \pi(a_t | s_t) \quad (\text{actor})$$

$$e(t) = \gamma \lambda \cdot e(t-1) + \nabla_W V(s_t) \quad (\text{critic})$$

where $\nabla_W \log \pi(a|s)$ encodes *which action was chosen* and in which direction to strengthen the weights to increase its probability.

Biologically, the eligibility trace corresponds to what Frémaux & Gerstner [3] call a **synaptic flag**: a pre/post-synaptic co-activation leaves a transient molecular trace at the synapse, without immediately triggering a weight change. This candidate for synaptic change only consolidates into a lasting modification if a neuromodulatory signal (here dopamine encoding δ) arrives within the appropriate time window. Without this third signal, the trace decays without leaving any lasting effect.

Since the cortical representation evolves throughout training, we chose to concatenate it with the raw observations before feeding them to the basal ganglia and the cerebellum. This provides a stable foundation that helps stabilize training without requiring a cortex warm-up phase. This could correspond biologically to sensory projections directly received by the thalamus before being sent to the basal ganglia.

7.2 Conceptual critiques and improvements

Beyond technical corrections, Doya’s model raises several fundamental criticisms: we present three of them, but only one of these three proposals will be implemented.

From passive cortex to predictive coding. In Doya’s theory, the cortex is a purely reactive feature extractor: it receives an input x , computes a compressed representation y , and does nothing more. No top-down signal modulates this representation based on what the rest of the brain expects. However, neocortical physiology also includes descending connections (which are even more numerous than ascending ones!). The **predictive coding** theory developed by Friston 2005 [4], based on the work of Rao and Ballard 1999 [10], proposes that the brain is constantly generating top-down predictions about its sensory inputs, and that only the *prediction error*—the difference between what is expected and what is observed—is propagated upward. Friston shows that minimizing this prediction error is equivalent to minimizing variational free energy. This architecture is both more computationally efficient and more physiologically consistent. Implementing it here would require transforming the cortex into a hierarchical generative model, which would require completely reworking our existing architecture.

Blurring the boundaries between learning paradigms. The strict separation of learning processes in Doya’s model is conceptually very elegant, but represents a simplification that is now known to be physiologically inaccurate. Indeed, we now know that dopaminergic axons project to the prefrontal cortex via the mesocortical pathway, as well as to the cerebellum, where traces of these signals have been observed in Purkinje cells. A relevant improvement would therefore be to introduce

this dopamine permeability, using it as a global modulatory signal via the three-factor rule that we have already implemented for the basal ganglia MLPs.

Towards spiking neural networks and STDP. The deepest limitation of the model lies in its level of description: all signals are firing rates, and all plasticity rules operate on temporal averages. However, biological neurons communicate through discrete spike trains, and synaptic plasticity critically depends on the precise timing between pre- and post-synaptic action potentials (STDP). This mechanism has conceptually different properties from our Hebbian rules: it induces LTD or LTP depending on the sign of the temporal shift, and is highly sensitive to the precise sequence of events. Implementing Doya’s model using **spiking neural networks** (*SNN*) would achieve a much higher level of biological plausibility, at the cost of considerable learning instability: non-differentiable signals make reinforcement learning in SNNs still a largely open problem.

Given the improvement opportunities of these three directions, and their implementation difficulty within the current architecture, we chose to implement direction #2: adding a modulatory dopaminergic signal in the cortex and cerebellum. This option provides the best compromise: it is biologically well motivated, integrates naturally into the existing architecture via the three-factor rule already implemented for the basal ganglia, and does not require a complete redesign of the existing modules.

7.2.1 Implementation

The `CerebralCortex3F` and `Cerebellum3F` modules extend the base modules by introducing the dopaminergic signal δ as a third factor of plasticity. For the cortex, the rule becomes:

$$\Delta W(t) = \alpha \cdot \delta_{\text{centered}}(t) \cdot e(t-1), \quad e(t) = y_t \otimes (x_t - W^\top y_t)$$

where $\delta_{\text{centered}} = \delta - \hat{\delta}$ is centered around an EMA $\hat{\delta}$ to ensure bidirectional modulation: thus the cortex consolidates its representations only when the current situation turns out to be better than expected. It therefore learns to represent states that lead to good surprises, rather than all states indiscriminately. A one-step delay separates the computation of $e(t)$ (available from `encode_and_learn`) from the arrival of δ_t (returned by `bg.learn`): the trace is therefore stored in `_pending_dW` and applied at the next step, which can be justified by the biological delay in dopamine diffusion.

For the cerebellum (`Cerebellum3F`), δ does not modulate the Hebbian rule but directly the LTD learning rate via a bounded sigmoid gate:

$$\eta_{\text{eff}} = \eta \cdot [1 + s \cdot (\sigma(\delta_{\text{centered}}) - 0.5)]$$

Thus, in the cerebellum, dopaminergic modulation allows transitions leading to good rewards to be learned faster than neutral or negative transitions, which accelerates the construction of the forward model on effective behaviors.

8 Evaluation of technical and conceptual improvements

8.1 Protocol

The new architectures are also evaluated on **CartPole-v1**. The protocol is strictly identical to that used in Section 6, but this time involves:

- **2F**: the architecture with the improved cortex and basal ganglia MLP + three-factor rule.
- **3F-Cortex**: the same architecture, but with the three-factor rule with dopamine in the cortex.
- **3F-Cerebellum**: the same architecture, but with the three-factor rule with dopamine in the cerebellum.
- **3F-Full**: the same architecture, but with the three-factor rule with dopamine in the cerebellum.

8.2 Results

The results are presented in Figure 8.

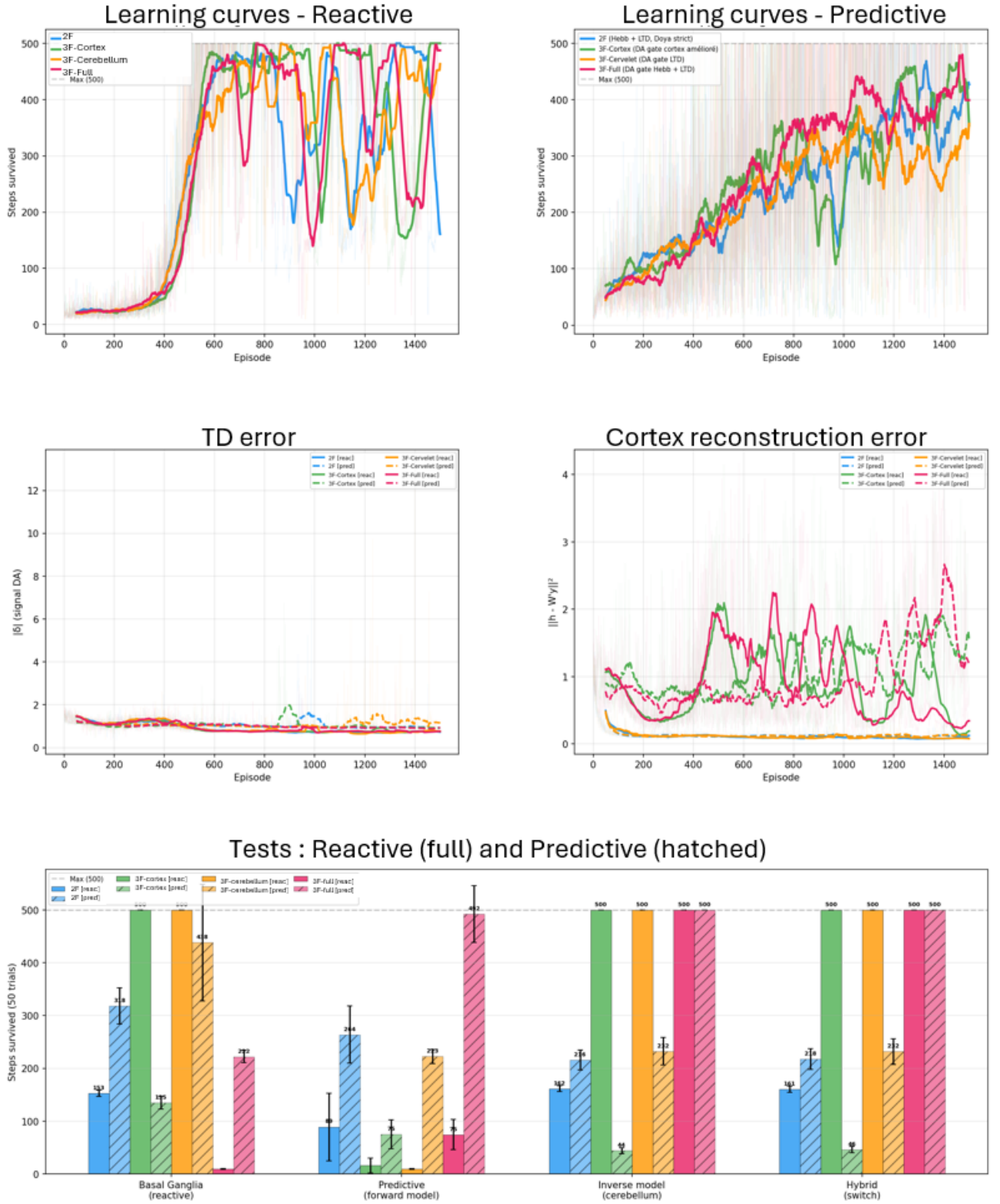


Figure 8: Benchmark — Improved Architectures on CartPole-v1: Two-Factor Baseline (2F) vs. Three-Factor Dopamine-Gated Variants (3F-Cortex, 3F-Cerebellum, 3F-Full), Reactive and Predictive Modes

In reactive mode, the learning curve follows a sigmoid, taking off between epochs 400 and 450 to reach a score above 450 at epoch 600. The curves then exhibit large oscillations corresponding to the exploratory policy during training. In predictive mode, the learning curves show a slower and more linear progression. All models reach a score above 350 by epoch 1500, but none appears to have reached a plateau. The 3F-Full variant achieves the best rewards at the end of training, with a peak at 480 around epoch 1450.

The TD signal (bottom-left panel) remains stable and of low amplitude for all variants ($|\delta| < 2$), confirming that the

modifications did not destabilize the dopaminergic signal. However, the cortical reconstruction error is higher for 3F-Cortex and 3F-Full, especially in predictive mode.

In the test phase, the results strongly depend on the strategy. For the *Basal Ganglia* strategy, 3F-Cortex and 3F-Cerebellum in reactive mode reach 500 steps (perfect score), compared to 150 steps for 2F. There seems to be an implementation issue with the reactive mode of 3F-Full. In predictive mode, 2F rises to 320 steps, while 3F-Cortex drops to 140 steps. 3F-Cerebellum remains the best (438 steps but with high variability), despite a slight decrease compared to reactive mode.

For the predictive strategy, training in reactive mode naturally leads to poor results—the basal ganglia were not trained for planning. However, one can note the very good score of 3F-Full in predictive mode: 492 steps, close to the perfect score with low variance. The performance of all other models in predictive mode is worse than before.

The Inverse model and Hybrid strategies present almost identical metrics: the model switches to basal ganglia mode only when $\|x_{\text{obs}} - x_{\text{predicted}}\|$ exceeds 2 standard deviations. Since learning has been effective, the model remains almost always in Inverse mode, which explains why the results are so similar. It can also be observed that all 3F variants in predictive mode reach the perfect score of 500 steps with near-zero variance. It is also noteworthy that in predictive mode, the 3F-Full model achieves a perfect score of 500 steps, which none of the other architectures achieves after this type of training.

8.3 Discussion

The technical improvement is validated. The transition to the improved cortex, combined with MLPs with eligibility traces for the basal ganglia, resolves the two structural limitations identified previously. The cortical representation is now sufficiently rich for the basal ganglia to extract a viable policy, as evidenced by the onset of learning within the first 400–500 episodes—where the original architecture stagnated throughout training.

Adding dopamine in the cortex improves test performance at the cost of less faithful reconstruction. The increase in reconstruction error for 3F-Cortex is not a flaw: it reflects exactly the expected effect of the three-factor rule. The cortex no longer learns to faithfully represent *all* states, but to over-represent states associated with positive surprises. This functional specialization translates into better test performance for strategies that rely on cortical representation (BG, Hybrid), at the expense of pure reconstruction. However, this architecture seems to make inference by the basal ganglia more difficult: by biasing the cortical representation toward rewarded states, the coverage of the state space required for good predictive evaluation is reduced.

Dopaminergic modulation of the cerebellum helps in predictive mode. Similarly, adding dopamine to the cerebellum improves model performance in both reactive and predictive learning modes: accelerating LTD on rewarded transitions makes the forward model more accurate where it matters, improving the quality of predictions used for action selection.

Double modulation (3F-Full) combines the strengths of 3F-Cortex and 3F-Cerebellum. The perfect scores after both reactive and predictive training show that the cerebellum perfectly encapsulates the best basal ganglia strategy. This is likely due to a strong synergy between the two modulations: the cortex provides a representation biased toward rewarded states, guiding basal ganglia exploration toward promising transitions, while dopaminergic modulation in the cerebellum amplifies LTD precisely on those same transitions—the two effects reinforce each other to accelerate the encapsulation of the

optimal policy in the inverse model. However, the Basal Ganglia strategy alone after predictive training seems to inherit the drawbacks introduced by 3F-Cortex.

9 Ablation experiments and simulation of neuropathologies

Ablation experiments and neuropathology simulations aim to provide a first level of physiological validation of the model. By removing or degrading a given module, one should observe specific and biologically predictable deficits. Although these experiments are a strong simplification of reality and cannot replace validation against physiological data, but they still offer a useful qualitative benchmark.

Three ablation experiments were conducted, along with two simulated pathologies (Parkinson’s disease and cerebellar ataxia). Only the observations are reported in this section. The corresponding figures are provided in the appendix.

9.0.1 Cortical module ablation

- **Modification:** The cortex is bypassed: the basal ganglia receive only the raw state $s \in \mathbb{R}^6$. The model is then trained for 800 epochs in predictive mode.
- **Prediction:** Slower convergence due to a less suitable representation. However, the basal ganglia retain their own nonlinearities (via MLPs), which may partially compensate.
- **Biological relevance:** Analogous to a partial cortical lesion where sensory representations are degraded but subcortical evaluation mechanisms remain preserved [14].
- **Result:** As expected, convergence is slower, but the basal ganglia remain sufficiently powerful to solve the task without cortical input thanks to their intrinsic nonlinear processing (see Appendix Fig. 1).

9.0.2 Forward model ablation

- **Modification:** The cerebellar prediction function is disabled. The model is trained in predictive mode for 800 epochs, followed by a test phase (50 trials $\times$ 4 conditions).
- **Prediction:** The predictive strategy should collapse, confirming that its advantage relies entirely on the forward model. Encapsulation should remain unaffected. No direct biological study was identified; this experiment primarily serves as a computational validation.
- **Result:** The predictive strategy is completely degraded: the agent fails to survive more than 9 steps. No impact is observed on the inverse mode. This confirms the functional dissociation between planning (forward model) and automatic control (inverse model) (see Appendix Fig. 2).

9.0.3 Inverse model ablation

- **Modification:** Encapsulation is disabled; the basal ganglia policy is never transferred to the cerebellum. Training is performed for 800 epochs followed by testing (50 trials $\times$ 4 conditions).

- **Prediction:** The inverse strategy collapses, with little to no impact on training or on basal ganglia and predictive strategies.
- **Biological relevance:** Simulates the inability to automatize learned behaviors: the agent “knows” how to perform the task but cannot execute it automatically. This is analogous to certain cerebellar patients who can voluntarily plan movements but lose motor fluency [12].
- **Result:** Learning is slightly slower, and the inverse strategy fails (less than 9 steps), as predicted. Since training was conducted in predictive mode, it is expected that the basal ganglia-only (reactive) strategy performs poorly. However, an unexpected collapse of the predictive strategy is observed: because the inverse model never learned to imitate the basal ganglia policy, the teaching signal provided to the forward model is degraded, which indirectly reduces prediction quality and harms action selection.

9.0.4 Simulation of Parkinson’s disease

Biological mechanism: Parkinson’s disease results from the progressive degeneration of dopaminergic neurons in the substantia nigra pars compacta (SNc), leading to a reduced TD signal δ and a drastic decrease in corticostriatal plasticity.

Protocol: Three stages of progression are simulated: Stage 1 (mild, -25%): $\delta \times 0.75$; Stage 2: $\delta \times 0.40$; Stage 3: $\delta \times 0.40$. Training is performed for 800 epochs, followed by testing over 50 trials under BG and Inverse conditions, with encapsulation performed either before or after training.

Results and discussion:

A progressive slowdown of learning is observed as Parkinsonian severity increases, eventually preventing the agent from learning the task (see Appendix Fig. 1). However, if cerebellar encapsulation occurs before the lesion, the cerebellum maintains acceptable performance even without dopamine (see Appendix Fig. 3).

This result is physiologically incomplete: although Parkinson’s patients can retain overlearned automatic behaviors, basal ganglia circuits remain essential for movement initiation, which can still be impaired. This symptom is absent in our model, revealing an architectural limitation: dopamine is modeled only as a scalar modulator of plasticity, capturing its role in learning but not in real-time motor control. In reality, dopamine dynamically regulates the balance between the direct (facilitatory) and indirect (inhibitory) pathways in the basal ganglia, a mechanism central to Parkinsonian motor deficits [6].

Additionally, tremor and rigidity are not captured by the model, as they arise from oscillatory dynamics in cortico-basal-thalamo-cortical loops, which are beyond the scope of this framework.

9.0.5 Simulation of cerebellar ataxia

Biological mechanism: Cerebellar ataxia is characterized by motor inaccuracy (dysmetria), intention tremor, and an inability to finely tune movements. Patients can initiate movements but fail to reach targets accurately [9].

Protocol: Ataxia is simulated *at test time* (`ataxia_test`) on a previously trained intact model, without retraining. Gaussian noise $\mathcal{N}(0, \sigma^2)$ with $\sigma = 0.5$ is added directly to Purkinje cell weights (`pc_weights`).

This simulates degradation of parallel fibers, introducing imprecision in forward predictions and inverse motor commands, analogous to clinically observed dysmetria.

Prediction: Both predictive and inverse strategies should degrade (due to corrupted forward predictions and imprecise motor commands), while the reactive basal ganglia strategy should remain intact, confirming the functional dissociation between basal ganglia and cerebellum.

Result: The results match the prediction. The hybrid strategy is also strongly degraded: the `HybridController` relies on the forward model, which is heavily perturbed by noise, leading to incorrect switching decisions. Additionally, inverse model impairments further contribute to degraded performance.

10 Learning a high-level task: Super Mario Bros

Our 3F-Full model achieved excellent results on CartPole. In this section, we challenge it on a significantly more complex task: playing *Super Mario Bros*. This task is fundamentally different: the cortex no longer receives a 4-dimensional state vector, but a sequence of pixel frames of size $84 \times 84 \times 4$, from which it must learn to identify relevant entities (Mario, enemies, obstacles, coins) and extract a representation suitable for control.

10.1 Architecture and implementation

The overall architecture is orchestrated by the `MarioBrain` module, which connects the three components of the 3F-Full architecture.

switching to a CNN in the cortex Since the input is now an image, the `CerebralCortexImproved` module can no longer be used directly. It is replaced by a NatureCNN-style architecture [7] (`VisualCortexCNN`), whose encoder follows:

$$(4 \times 84 \times 84) \xrightarrow{\text{Conv}_{8,s4}} \xrightarrow{\text{Conv}_{4,s2}} \xrightarrow{\text{Conv}_{3,s1}} \xrightarrow{\text{Flatten}} \xrightarrow{\text{FC}_{512}} \xrightarrow{\text{FC}_{256}} \mathbf{z} \in \mathbb{R}^{256}$$

This architecture is not so different from `CerebralCortexImproved`: convolutional layers play the role of nonlinear expansion, while fully connected layers correspond to a learned sparse encoding.

To retain the reconstruction objective $\|x - W^\top y\|^2$, we add a decoding head (`fc_decoder` + `decoder_conv`) capable of reconstructing the input. The dopaminergic signal is incorporated directly into the effective learning rate:

$$\eta_{\text{eff}} = \eta \cdot [1 + s \cdot (\sigma(\delta_{\text{centered}}) - 0.5)]$$

However, to fully leverage this visual cortex, we use backpropagation with the Adam optimizer to update weights. While this reduces biological plausibility, it provides the most stable learning (predictive coding could be an alternative, but remains harder to stabilize in this setting, and remains more complicated to implement).

Basal ganglia and cerebellum. The basal ganglia (`BasalGangliaMario`) retain the same MLP architecture with $TD(\lambda)$ eligibility traces developed for CartPole, with a 256-dimensional input (visual features) and 7 output actions. The softmax temperature is linearly annealed from 2.5 to 0.3 over 800 epochs to gradually shift from exploration to exploitation. The reward signal include The reward signal combines extrinsic environment feedback, a progress bonus proportional to horizontal displacement Δx , three penalty terms for immobility, backward movement, and life loss and an intrinsic curiosity term (ICM)

[8] :

$$r_{\text{shaped}} = r_{\text{ext}} + r_{\text{ICM}} + r_{\text{progress}} - p_{\text{immobility}} - p_{\text{death}} - p_{\text{backward}}$$

The cerebellum is adapted to the continuous feature space: `ForwardModelNeural` and `InverseModelNeural` replace granule cells with MLPs trained via backpropagation, while preserving dopaminergic modulation of the learning rate via the same sigmoid gate. The forward model predicts $\mathbf{z}_{t+1}$ from $(\mathbf{z}_t, a_t)$ (with a_t one-hot encoded), while the inverse model predicts a_t from $(\mathbf{z}_t, \mathbf{z}_{t+1})$.

These models are integrated into an Intrinsic Curiosity Module [8] : the forward model prediction error serves as a normalized intrinsic reward, encouraging exploration of visually novel transitions—crucial in sparse-reward environments like Mario. The inverse model error regularizes the feature space to encode only control-relevant information rather than stochastic visual noise.

10.2 Results and discussion

After one day of training, the agent is able to complete the first level (`SuperMarioBros-1-1`). This performance is not comparable to what we obtained with a task-specific architecture (CNN + PPO : level 1-1 completed in less than 7 hours), but it demonstrates the strong capabilities of the model. However, the use of backpropagation and Adam for updating the CNN and cerebellar MLPs significantly reduces biological plausibility.

This experiment highlights a key trade-off: as task complexity increases, it becomes increasingly difficult to maintain both biological plausibility and high performance.

11 Conclusion and outlook

To our knowledge, this work provides the first complete implementation of Doya’s tripartite framework (1999, 2000): unsupervised cortex, reinforcement-learning basal ganglia, and supervised cerebellum, all operating simultaneously on a shared task with their original learning principles.

Beyond this reference implementation, we introduced and evaluated a series of technical and conceptual improvements extending Doya’s theory in biologically motivated directions. We also demonstrated that the architecture can scale to complex tasks such as Super Mario Bros, albeit at the cost of reduced biological plausibility in some components. The best trade-off between plausibility and performance is achieved by the 3F-Full architecture, combining MLP-based basal ganglia, an improved multilayer cortex, and the original cerebellar module. This approach departs from Doya’s strict separation of learning paradigms: in the 3F-Full model, dopamine modulates all networks. As a result, cortical unsupervised learning becomes reward-biased representation learning, while cerebellar supervised learning becomes performance-weighted adaptation.

This work highlights a fundamental tension: biological plausibility and empirical performance often pull in opposite directions. It also suggests that the core challenge of computational neuroscience is not merely to reproduce machine learning performance with biological constraints, but to understand why the brain achieves such efficiency despite (or because of !) those constraints.

Several natural directions emerge from this work. First, implementing predictive coding in the cortex would be a major step forward, potentially replacing backpropagation and Adam while significantly improving biological plausibility. It would be particularly interesting to study how predictive coding interacts with the three-factor learning rule already implemented in the cortex. Second, moving toward spiking neural networks (SNNs) with spike-timing-dependent plasticity (STDP) would further increase physiological realism, though at the cost of substantial learning instability. Finally—and most importantly—the model should be rigorously validated against real electrophysiological data.

References

- [1] Daniele Caligiore, Giovanni Pezzulo, et al. “Consensus Paper: Towards a Systems-Level View of Cerebellar Function: the Interplay Between Cerebellum, Basal Ganglia, and Cortex”. In: *Cerebellum (London, England)* 16.1 (2017), pp. 203–229. DOI: 10.1007/s12311-016-0763-3. URL: <https://doi.org/10.1007/s12311-016-0763-3>.
- [2] Kenji Doya. “What are the computations of the cerebellum, the basal ganglia and the cerebral cortex?” In: *Neural Networks* 12.7–8 (1999), pp. 961–974.
- [3] Nicolas Frémaux and Wulfram Gerstner. “Neuromodulated Spike-Timing-Dependent Plasticity, and Theory of Three-Factor Learning Rules”. In: *Frontiers in Neural Circuits* 9 (2016), p. 85. DOI: 10.3389/fncir.2015.00085. URL: <https://doi.org/10.3389/fncir.2015.00085>.
- [4] Karl Friston. “A Theory of Cortical Responses”. In: *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences* 360.1456 (2005), pp. 815–836. DOI: 10.1098/rstb.2005.1622. URL: <https://doi.org/10.1098/rstb.2005.1622>.
- [5] Joaquin M. Fuster. *The Prefrontal Cortex*. 4th. Consulté le 15 avril 2026. Academic Press, 2008. ISBN: 978-0123736444. URL: <https://www.sciencedirect.com/book/9780123736444/the-prefrontal-cortex>.
- [6] Francesca Magrinelli et al. “Pathophysiology of Motor Dysfunction in Parkinson’s Disease as the Rationale for Drug Treatment and Rehabilitation”. In: *Parkinson’s Disease* 2016 (2016), p. 9832839. DOI: 10.1155/2016/9832839. URL: <https://doi.org/10.1155/2016/9832839>.
- [7] Volodymyr Mnih et al. “Human-level control through deep reinforcement learning”. In: *Nature* 518.7540 (2015), pp. 529–533. DOI: 10.1038/nature14236. URL: <https://doi.org/10.1038/nature14236>.
- [8] Deepak Pathak et al. “Curiosity-driven Exploration by Self-supervised Prediction”. In: *ICML*. 2017.
- [9] Sara Radmard, Ricardo Chaves, and Tetsuo Ashizawa. “Evaluation of Cerebellar Ataxic Patients”. In: *Neurologic Clinics* 41.1 (Feb. 2023), pp. 21–44. DOI: 10.1016/j.ncl.2022.05.002. URL: <https://doi.org/10.1016/j.ncl.2022.05.002>.
- [10] Rajesh P. N. Rao and Dana H. Ballard. “Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects”. In: *Nature Neuroscience* 2.1 (1999), pp. 79–87.
- [11] Wolfram Schultz, Peter Dayan, and P. Read Montague. “A neural substrate of prediction and reward”. In: *Science* 275.5306 (1997), pp. 1593–1599.

- [12] Ayesha R. Thanawalla et al. “The Cerebellar Nuclei and Dexterous Limb Movements”. In: *Neuroscience*. A commemoration of Thomas M. Jessell’s Contributions to Neuroscience 450 (2020), pp. 168–183. DOI: 10.1016/j.neuroscience.2020.06.046. URL: <https://doi.org/10.1016/j.neuroscience.2020.06.046>.
- [13] Daniel M Wolpert, R Chris Miall, and Mitsuo Kawato. “Internal models in the cerebellum”. In: *Trends in cognitive sciences* 2.9 (1998), pp. 338–347.
- [14] Lusha Zhu et al. “Patients with basal ganglia damage show preserved learning in an economic game”. In: *Nature Communications* 10.1 (Feb. 2019), p. 802. DOI: 10.1038/s41467-019-08766-1. URL: <https://doi.org/10.1038/s41467-019-08766-1>.